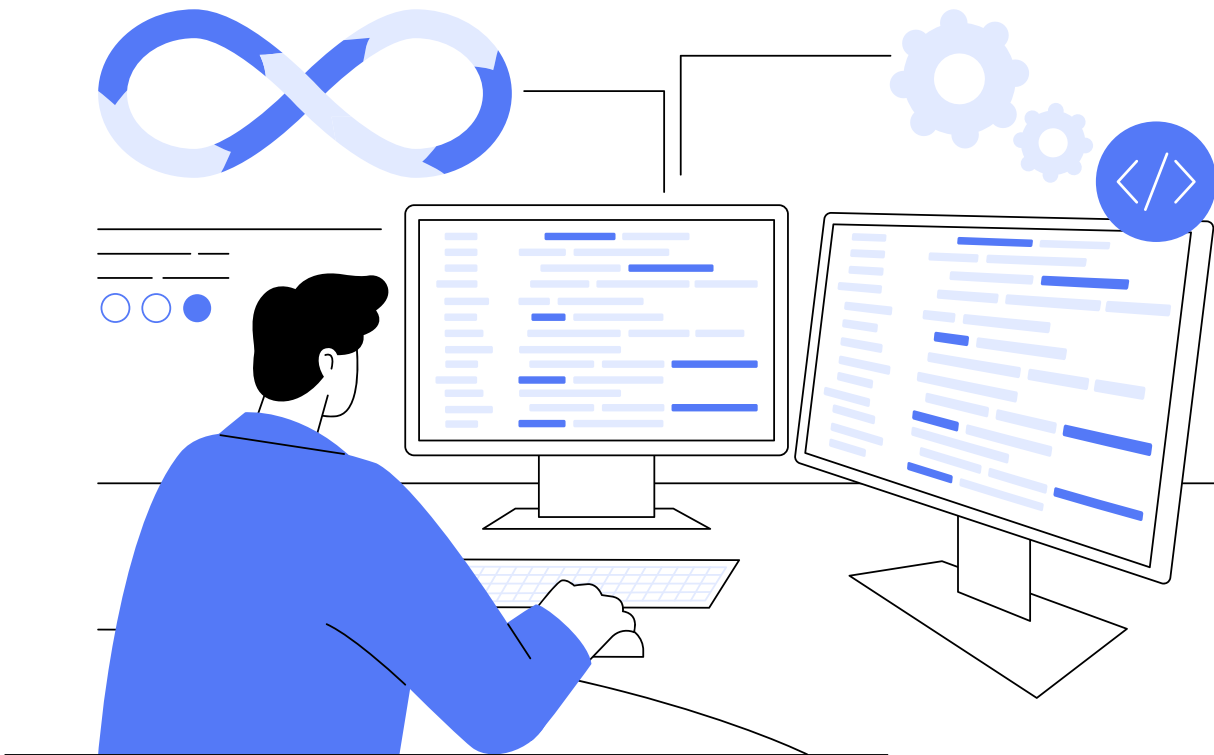


Observability: Indispensable for Modern DevOps

Observability helps teams detect failures and gain a deep understanding of their root causes. This simplifies debugging while improving systems performance and reliability. Modern DevOps benefits greatly from open source observability tools.



In the past, IT teams depended on monitoring systems to check system functionality. Monitoring tools generated alerts when system failures occurred, such as server crashes or high CPU usage. These tools allowed users to track basic system metrics such as memory utilisation, system errors or system load. This approach worked well when applications were simple and ran on a few servers.

But modern systems are much more complex. The current technology landscape employs microservices together with containers and cloud platforms. Many applications are now built using microservices architecture, where each feature like user login, payment processing, or order tracking is handled by its own independent service. These microservices exchange information through APIs while operating from different servers and containers across multiple cloud provider platforms. For example, in an ecommerce application, the product catalogue may run on AWS, the payment service

on Azure, and the customer support chatbot may run on-premises. These different system components need to function together to provide users with a seamless experience. Such complex systems require more than a fundamental understanding of system failures. We need to know what broke, where it broke, and why it broke.

The solution to this problem is observability. Observability extends beyond traditional monitoring capabilities by enabling teams to understand what's happening inside complex systems even when unexpected issues occur. It uses logs, metrics and traces to deliver complete insights on system behaviour.

Observability functions as a control room in the fast-moving DevOps environments that experience continuous software changes. It helps teams to detect the root causes of problems so they can resolve them swiftly while developing more reliable systems. Observability provides teams with clarity and confidence. Without it, teams lack a clear direction.